Undergraduate seminar: Counting collisions and measuring π.

Oleg Starykh

Professor
Department of Physics and Astronomy
University of Utah
Salt Lake City, UT 84112 - 0830

Office: JFB 304
Phone: 801-581-6424
Fax: 801-581-4801
Email: **starykh at physics.utah.edu**

I am condensed matter theorist. I am studying quantum phases of strongly correlated electron systems, quantum phase transitions between them and, in general, competition between classical and quantum ordering tendencies.

Teaching: course pages and materials

Research: recent talks and publications

Google Scholar page

Condensed Matter Seminar

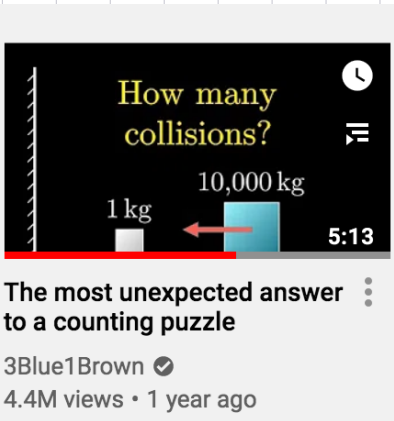
Few random curiosities

I am condensed matter theorist studying quantum magnetism, superconductivity and many other interesting problems.

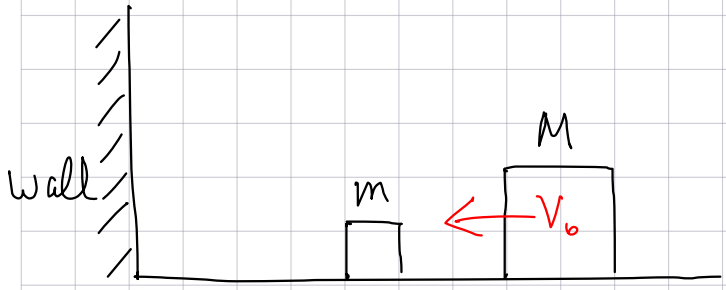
But today I want to describe to you a rather simple, but very unexpected, problem which you can solve by very elementary means — only using energy and momentum conservation, and some math tricks.

The problem is about an unexpected connection between colliding blocks and the number $\pi \approx 3.14159265\ldots$

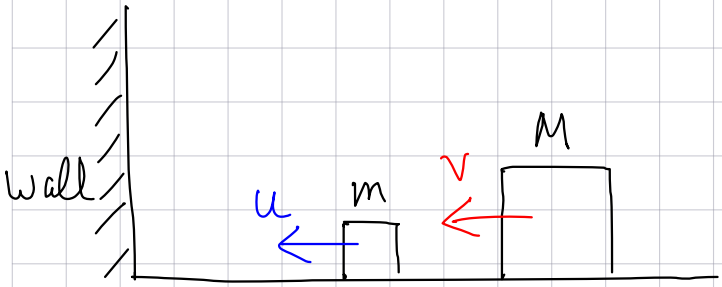
It comes on the website 3Blue1Brown which I highly recommend



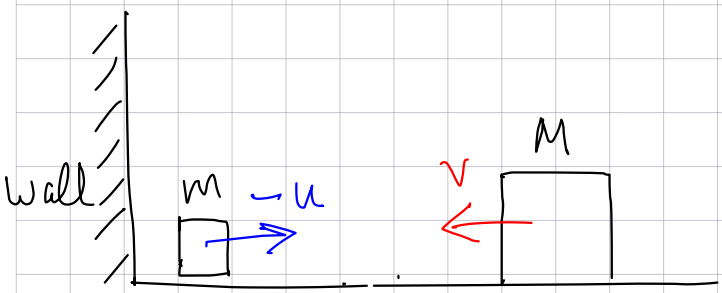
The problem is formulated like this: two blocks, of mass m and M, are free to slide without friction along the horizontal surface.



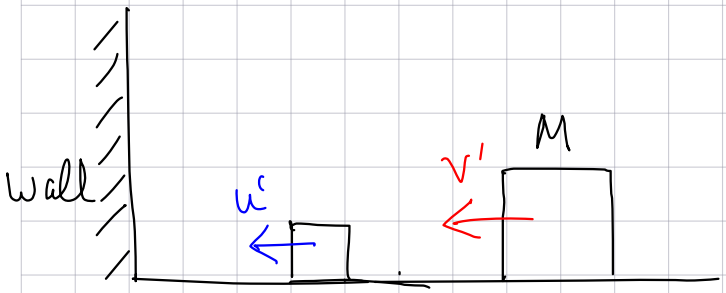
Block M is given initial velocity V_0, sliding to the left towards block m and the wall.



It hits block m which too begins to slide with some velocity u.



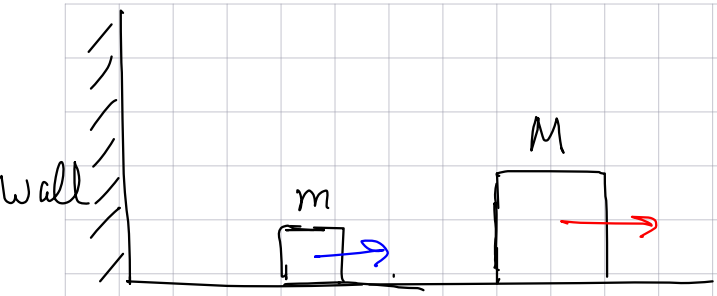
Block m hits the wall and reflect back with velocity $-u$: the collision with the wall (which is infinitely massive) is elastic



The two blocks collide again, acquiring velocities V' and u', and the process continues.

All collisions are elastic!

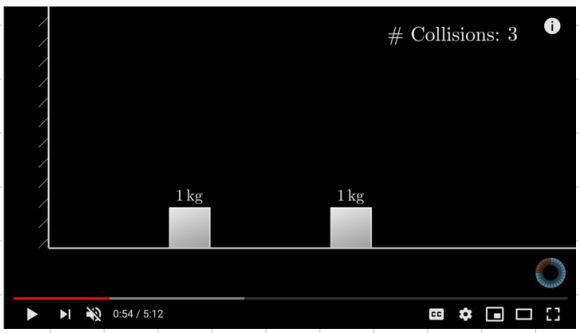
always!!!



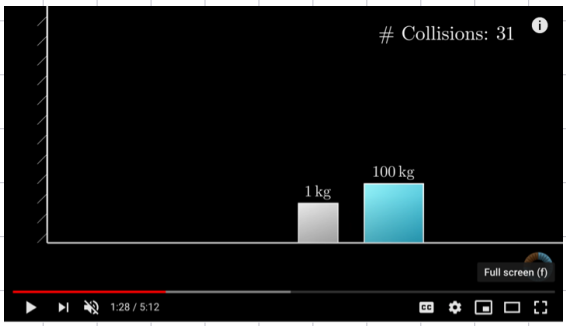
Eventually, block M too collides with the wall (squeezing block m between the wall and itself) and elastically reflects back, moving out to the positive infinity.

The question is: how many collisions the smaller block (of mass m) experiences?

The answer is quite surprising. The number of collisions N_{tot} depends on the ratio $\boxed{r = \frac{M}{m}}$ but does not depend on V_0.



For $r = \frac{M}{m} = 1$ $N_{tot} = 3$



For $r = 100$ $N_{tot} = 31$

And for $r = 10000 = 10^4$

$N_{tot} = 314$.

For $r = 10^8$ $N_{tot} = 31415$ collisions... And so on.

So clearly N_{tot} is given by digits of the number π.

But why?

We will solve this now.

So, we have quite a surprising observation. Let us now get to work and figure out what is going on.

Step 1: analyze single collision between blocks m and M with initial velocities u and v.

Before:

After:

Momentum is conserved: $mu + Mv = mu' + Mv'$ (1)

Collisions are elastic, so the energy is also conserved: $\frac{1}{2}mu^2 + \frac{1}{2}Mv^2 = \frac{1}{2}m(u')^2 + \frac{1}{2}M(v')^2$ (2)

Note: the motion is one-dimensional and therefore
$u > 0$ means motion to the right,
$u < 0$ means motion to the left.

Re-arrange equations (1) and (2):

$$\begin{cases} m(u-u') = M(v'-v) & (1') \\ m\left(u^2-(u')^2\right) = M\left((v')^2-v^2\right) & (2') \end{cases}$$

Use $a^2-b^2 = (a-b)(a+b)$ to rewrite (2') as

$$\underbrace{m(u-u')}_{\text{equal by eq. (1')}}(u+u') = \underbrace{M(v'-v)}_{\text{equal by eq. (1')}}(v'+v)$$

So we obtain $u+u' = v'+v$ (2'')

$\Rightarrow u' = v'+v-u$ and use this in (1')

Then

$$m\left(u-(v'+v-u)\right) = M(v'-v)\text{, and re-arrange}$$

$$m(2u-v)+Mv = (m+M)v'$$

to find $$v' = \frac{(M-m)v+2mu}{M+m} \qquad (3)$$

and therefore

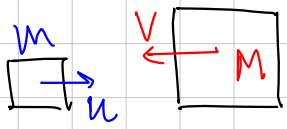
$$u' = v'+v-u = \frac{(M+m)(v-u)+(M-m)v+2mu}{M+m} \quad \text{so that}$$

$$u' = \frac{(m-M)u+2Mv}{M+m} \qquad (4)$$

Step 2: small block m next collides elastically with the wall and reverses the direction of its motion

$\Rightarrow$ its velocity after the collision with the wall is just $$u'' = -u' = \frac{(M-m)u-2Mv}{M+m} \qquad (5)$$

Thus, if before Step 1 we had



Then after Step 2 we have

$$u'' = \frac{(M-m)u-2Mv}{M+m} \qquad (5)$$

$$v' = \frac{(M-m)v+2mu}{M+m} \qquad (3)$$

Now, we can generalize: after n-th collision block m has velocity u_n and block M has velocity V_n.

Then after going through $(n+1)$-th collision, <u>and</u> after block m colliding with the wall, they will have velocities u_{n+1} and V_{n+1} correspondingly.

By virtue of equations (5) and (3) we know that

$$(6)\begin{cases} u_{n+1} = \dfrac{(M-m)u_n - 2MV_n}{M+m} \\ V_{n+1} = \dfrac{(M-m)V_n + 2mu_n}{M+m} \end{cases}$$

. Use $M = rm$ $(r = M/m)$ to rewrite this as

$$(7)\begin{cases} u_{n+1} = \dfrac{(r-1)u_n - 2rV_n}{r+1} \\ V_{n+1} = \dfrac{(r-1)V_n + 2u_n}{r+1} \end{cases}$$

Since energy is conserved, $\mathcal{E} = \frac{1}{2}mu_n^2 + \frac{1}{2}MV_n^2 = \text{const}$

$\Rightarrow \dfrac{2\mathcal{E}}{m} = u_n^2 + \dfrac{M}{m}V_n^2 = u_n^2 + r^2V_n^2 = u_n^2 + (\sqrt{r}\,V_n)^2$ is the constant of motion. Multiply both sides of the 2nd equation in (7) by $\sqrt{r}$ to obtain:

$$(8)\begin{cases} u_{n+1} = \dfrac{(r-1)u_n - 2\sqrt{r}(\sqrt{r}\,V_n)}{r+1} \\ \sqrt{r}\,V_{n+1} = \dfrac{(r-1)(\sqrt{r}\,V_n) + 2\sqrt{r}\,u_n}{r+1} \end{cases}$$

Now, $\frac{2\mathcal{E}}{m} = u_n^2 + (\sqrt{r}\, v_n)^2 = \text{constant}.$

Step 3 We parameterize velocities by angle θ_n:

$$\begin{cases} u_n = \sqrt{\frac{2\mathcal{E}}{m}} \sin\theta_n \\ \sqrt{r}\, v_n = \sqrt{\frac{2\mathcal{E}}{m}} \cos\theta_n \end{cases} \qquad (9)$$

This is because $\frac{2\mathcal{E}}{m} = \frac{2\mathcal{E}}{m} \underbrace{(\sin^2\theta_n + \cos^2\theta_n)}_{1}$

Then
$$\begin{cases} u_{n+1} = \sqrt{\frac{2\mathcal{E}}{m}} \sin\theta_{n+1} \\ \sqrt{r}\, v_{n+1} = \sqrt{\frac{2\mathcal{E}}{m}} \cos\theta_{n+1} \end{cases}$$

Then equation (8) is re-written as recursive relation for angles θ_{n+1} and θ_n:

$$(10) \begin{cases} \sin\theta_{n+1} = \dfrac{(r-1)\sin\theta_n - 2\sqrt{r}\cos\theta_n}{r+1} \\ \cos\theta_{n+1} = \dfrac{(r-1)\cos\theta_n + 2\sqrt{r}\sin\theta_n}{r+1} \end{cases}$$

Step 4 We now use Euler's identity $\boxed{e^{i\alpha} = \cos\alpha + i\sin\alpha}$

where $i = \sqrt{-1}$

Use (10) to form $e^{i\theta_{n+1}} = \cos\theta_{n+1} + i\sin\theta_{n+1} =$

$$= \frac{(r-1)}{r+1}(\cos\theta_n + i\sin\theta_n) + \frac{2\sqrt{r}}{1+r}\underbrace{(\sin\theta_n - i\cos\theta_n)}_{(-i)(\cos\theta_n + i\sin\theta_n)} =$$

$$= \frac{r-1}{r+1} e^{i\theta_n} - i\frac{2\sqrt{r}}{1+r} e^{i\theta_n} = \frac{r-1-i2\sqrt{r}}{r+1} e^{i\theta_n}$$

We obtained a very compact relation:

$$\boxed{e^{i\theta_{n+1}} = \frac{r-1-i2\sqrt{r}}{r+1}\, e^{i\theta_n}} \qquad (11)$$

It remains to work out the coefficient, which is complex number, as $\frac{r-1-i2\sqrt{r}}{r+1} = \rho\, e^{-i\varphi}$

Its magnitude is $\rho = \left|\frac{r-1-i2\sqrt{r}}{r+1}\right| = \frac{\sqrt{(r-1)^2+(2\sqrt{r})^2}}{r+1} =$

$$= \frac{\sqrt{r^2-2r+1+4r}}{r+1} = \frac{\sqrt{r^2+2r+1}}{r+1} = 1$$

Its <u>real part</u> is: $\rho\cos\varphi = \frac{r-1}{r+1}$

Its <u>imaginary part</u>: $-\rho\sin\varphi = -\frac{2\sqrt{r}}{r+1}$

Therefore $\tan\varphi = \frac{\rho\sin\varphi}{\rho\cos\varphi} = \frac{2\sqrt{r}}{r-1}$.

Hence $\rho\, e^{-i\varphi} = \frac{r-1-i2\sqrt{r}}{r+1} = e^{-i\varphi}$,

$$\varphi = \arctan\left(\frac{2\sqrt{r}}{r-1}\right),$$

And thus equation (10) turns into:

$$e^{i\theta_{n+1}} = e^{-i\varphi}\, e^{i\theta_n}$$

$$e^{i\theta_{n+1}} = e^{i\theta_n - i\varphi} \qquad (12)$$

And $\boxed{\theta_{n+1} = \theta_n - \varphi}$ (13)

Equation (13) is remarkably simple, and yet it describes all collisions (actually, <u>double</u> collisions — collision between M and m, followed by the reflection of m off the wall).

$$\boxed{\theta_{n+1} = \theta_n - \varphi} \qquad (13)$$

For our problem, the initial condition is (see page 2)

$V_0 = -|V_0|$ (block M is sliding to the left) and $u_0 = 0$.

Then $\frac{2\mathcal{E}}{m} = u_0^2 + r V_0^2 = r V_0^2 \Rightarrow \sqrt{\frac{2\mathcal{E}}{m}} = \sqrt{r}\,|V_0|$.

Then
$$\begin{cases} u_n = \sqrt{r}\,|V_0| \sin\theta_n \\ \underbrace{\sqrt{r}\,V_n = \sqrt{r}\,|V_0| \cos\theta_n}_{V_n = |V_0|\cos\theta_n} \end{cases} \Rightarrow \begin{cases} u_0 = 0 = |V_0| \sin\theta_0 \Rightarrow \theta_0 = 0 \text{ or } \pi \\ \underbrace{\sqrt{r}\,V_0}_{-\sqrt{r}|V_0|} = \sqrt{r}\,|V_0| \cos\theta_0 \Rightarrow \theta_0 = \pi \end{cases}$$

Thus the initial angle is $\theta_0 = \pi$.

Then $\theta_{n+1} = \theta_n - \varphi$ means:

$$(14) \quad \begin{cases} \theta_1 = \theta_0 - \varphi = \pi - \varphi \\ \theta_2 = \theta_1 - \varphi = \pi - 2\varphi \\ \theta_3 = \pi - 3\varphi \\ \vdots \\ \theta_n = \pi - n\varphi \end{cases}$$

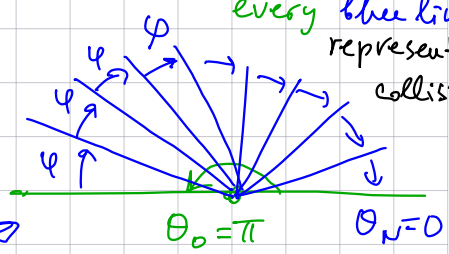
It remains to realize that collisions will stop when

$\theta_N = 0$ so that $V_N = |V_0| \cos\theta_N = |V_0|$

and $U_N = \sqrt{r}\,|V_0| \sin\theta_N = 0$

after the last collision:

Simple geometric picture every blue line represents collision.



Hence $\boxed{\theta_N = \pi - N\varphi} = 0 \Rightarrow \boxed{N = \frac{\pi}{\varphi}}$

Since $\varphi = \arctan\left(\frac{2\sqrt{r}}{r-1}\right)$, for large $r = \frac{M}{m} \gg 1$

it is convenient to write $r = \frac{1}{x}$ (so that $x = \frac{m}{M}$)

and expand $\varphi = \arctan\left[\frac{2}{\sqrt{x}} \frac{1}{\frac{1}{x} - 1}\right] = \arctan\left[\frac{2\sqrt{x}}{1-x}\right]$

in Taylor series in <u>small</u> $x = \frac{m}{M} \ll 1$.

Then $\varphi \simeq 2\sqrt{x} - \frac{2}{3} x^{3/2} \to 2\sqrt{\frac{m}{M}}$.

Hence $N = \frac{\pi}{2\sqrt{\frac{m}{M}}} = \frac{\pi}{2}\sqrt{\frac{M}{m}}$. Since each of the N events accounts for <u>2 collisions</u>, we obtain the main result:

$$\boxed{N_{tot} = 2N = \pi\sqrt{\frac{M}{m}}}$$

Therefore, if

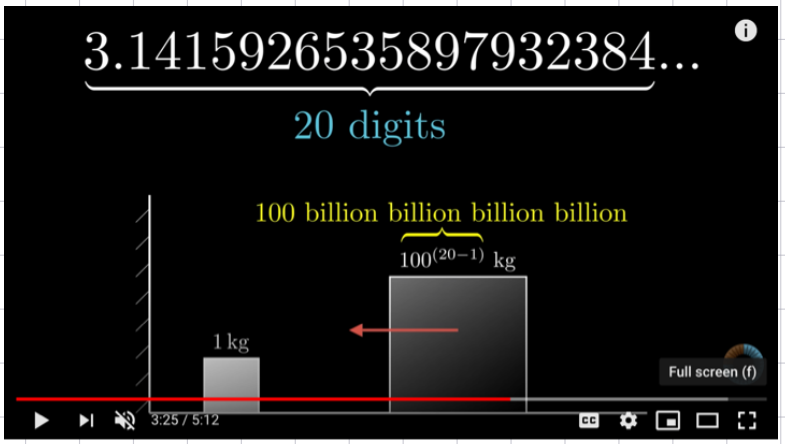
$\frac{M}{m} = 10^{2K}$, then $N_{tot} = \pi \cdot 10^K \Rightarrow$ the number of collisions is given by first K digits of the number π

where K = integer

Hope you find this problem interesting.
Its answer is certainly unusual. But observe that to solve it not much is required: we used conservation of the linear momentum and kinetic energy, and Euler's formula ($e^{ix} = \cos x + i \sin x$).

This problem also shows how theoretical physics works: you start with unusual observation and then formulate a simple model for it. The analysis of the model usually requires many mathematical steps. Provided that your original physical understanding was correct, you eventually solve the problem. (And if it was incorrect, you go back to the model formulation step and try to improve it.)

Do check out

3Blue1Brown.

They have many interesting problems and puzzles.